

EVO001BOT CRON PIPELINE PERSISTENCE HUNT

Linux process hunt for the source-observed
recurring downloader-to-shell command
pattern.

HUNTING QUERY

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceProcessEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

16 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceProcessEvents

ATT&CK

T1059.004 · T1053.003

Evidence-led dossier. Query status remains an untested
implementation sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Hunt the complete dual-downloader shell pipeline.

FortiGuard observed Evo001Bot installing a cron entry that executes every five minutes using `wget -q0-`, fallback `curl -sL`, a pipe into `/bin/sh` and output suppression. DeviceProcessEvents records the process command line on onboarded Linux servers.

OPERATIONAL VALUE

Selective execution evidence with explicit coverage limits.

The complete token combination is more selective than a generic curl-to-shell alert. Coverage is limited to Linux servers with Defender for Endpoint process telemetry; the edge devices most frequently targeted may not be onboarded.

8.8

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Hunting

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- Published 13 August 2026; active targeting observed since July 2026.
- The `!persist` command installs `systemd`, `SysV`, `cron`, `profile` and `rc.local` mechanisms.
- The cron entry runs the dual downloader-to-shell pipeline every five minutes.

ANALYTICAL ASSESSMENT

What the evidence supports

- The complete pipeline is a selective process-execution pivot.
- Full command-line visibility is essential.
- A match cannot prove cron ownership, successful download or attribution.

DETECTION ASSUMPTIONS

What must be validated

- The Linux server is onboarded to Defender for Endpoint.
- DeviceProcessEvents contains the complete `ProcessCommandLine`.
- Approved tooling does not reproduce the full source pattern.

Confidence boundary

FortiGuard provides malware and campaign telemetry, not tenant Defender events. The query is untested and does not prove successful download, script content, payload execution, persistence success, C2 traffic or Evo001Bot attribution.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Exploit edge device

Known vulnerabilities provide remote command execution.

02

Fetch loader

wget.sh selects one of twelve architecture-specific binaries.

03

Run bot

The binary performs analysis checks and connects on TCP/443.

04

Install persistence

!persist writes multiple startup mechanisms.

05

Repeat recovery

Cron executes the downloader pipeline every five minutes.

Observable core

A sh or bash process command line containing the complete wget/curl fallback pipeline, pipe to /bin/sh and output redirection.

Non-observable boundary

Cron-file ownership, HTTP response, downloaded script content, payload success, non-onboarded appliances, encrypted C2 content and attribution.

DETECTION HYPOTHESIS

Recurring persistence should execute a distinctive shell pipeline.

If Evooo1Bot executes the documented cron persistence, DeviceProcessEvents should record sh or bash with wget -qO-, fallback curl -sL, a pipe to /bin/sh and output redirected to /dev/null. Legitimate activity can be separated using cron origin, owner, URL, file and network evidence.

REQUIRED TELEMETRY

DeviceProcessEvents

- Microsoft Defender for Endpoint process telemetry.
- DeviceProcessEvents.
- FileName and full ProcessCommandLine.

OPTIONAL ENRICHMENT

Context that improves disposition

- Account and initiating-process context.
- Cron-file modification and owner.
- Destination URL, network event and downloaded-file hash.

UNAVAILABLE HERE

Do not infer

- Payload or HTTP-response content.
- Coverage for unmanaged routers and appliances.
- Proof of persistence or Evooo1Bot attribution.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR - DeviceProcessEvents

DeviceProcessEvents documents process creation, FileName, ProcessCommandLine, DeviceId, DeviceName and ReportId. Microsoft documents Linux onboarding requirements and the independent Defender telemetry pipeline.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

ReportId must be combined with DeviceName and Timestamp for event identity. No OSPlatform field, undocumented ActionType, parser assumption or command-success inference is used.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Process event	DeviceProcessEvents	Process creation and related activity.	REQUIRED
Executable	FileName	sh or bash.	REQUIRED
Command	ProcessCommandLine	Complete source-observed pipeline.	REQUIRED
Host	DeviceId, DeviceName	Asset pivot.	REQUIRED
Event identity	Timestamp, ReportId	Unique record pivot.	REQUIRED
Account	AccountName	Execution identity.	OPTIONAL
Parent	InitiatingProcessFileName, InitiatingProcessCommandLi ne	Cron or shell origin.	OPTIONAL
Path	FolderPath	Executable location.	OPTIONAL

Data quality requirements

- Confirm Linux device onboarding and sensor health.
- Verify full command-line population with a controlled command.
- Run split negative tests for fallback, pipe and redirection.
- Review historical collisions without a rarity threshold.

Known failure conditions

- Target device is not onboarded.
- ProcessCommandLine is absent or truncated.
- The actor changes downloader syntax or executes a binary directly.
- A global EDR exclusion suppresses telemetry.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

Find the source-observed recurring downloader pipeline.

Select only sh or bash executions containing all four FortiGuard-observed command fragments. Preserve full event and parent context for triage.

ANALYTICAL LOGIC

Direct filters and investigable output.

Executable filter plus exact command fragments. No IOC, hash, destination, join, sequence, aggregation, count, rarity, regex or automatic attribution.

KQL · UNTESTED HUNT

```
// Hunting query - Untested implementation sketch
// Primary platform: Microsoft Defender XDR / Advanced Hunting
// Required telemetry: Microsoft Defender for Endpoint process creation on Linux
// Evidence: FortiGuard observed Evooo1Bot installing a five-minute cron entry
// that executes this dual downloader-to-shell pipeline with output suppressed.
// A match is a persistence lead, not proof of Evooo1Bot attribution.
DeviceProcessEvents
| where FileName in~ ("sh", "bash")
| where ProcessCommandLine contains "wget -q0-"
| where ProcessCommandLine contains "|| curl -sL"
| where ProcessCommandLine contains "| /bin/sh"
| where ProcessCommandLine contains "> /dev/null 2>&1"
| project
    Timestamp,
    DeviceId,
    DeviceName,
    ReportId,
    FileName,
    FolderPath,
    ProcessCommandLine,
    AccountName,
    InitiatingProcessFileName,
    InitiatingProcessCommandLine,
    InitiatingProcessAccountName
| order by Timestamp desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunt	None	None	DeviceName + Timestamp + ReportId	Manual triage

What a match means

The host executed the complete Evooo1Bot-associated downloader-to-shell pattern and requires immediate cron, URL, process-tree and network review.

What a match does not prove

Cron ownership, successful download, script execution, persistence, C2 activity, compromise scope or Evooo1Bot attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify all command fragments, host, account, event identity and executable path.
- 02

Trace the parent.
Determine whether cron, init, profile or an interactive shell launched the process.
- 03

Inspect persistence.
Review crontabs, systemd, SysV, profile.d and rc.local changes.
- 04

Resolve the URL.
Collect destination, response metadata and downloaded bytes if available.
- 05

Scope the device.
Check binary hashes, port 443 traffic, SOCKS listeners and nearby exploit activity.

Escalation gate

Escalate when the full pipeline is unowned, originates from cron or startup files, reaches an unknown public destination, writes an executable, or appears with Evooo1Bot hashes, SOCKS activity, port 443 C2 or exploit evidence.

Primary benign overlap

Bootstrap installers and appliance management can use downloader-to-shell patterns. Tune only with verified owner, immutable script source, approved destination and change record; never suppress curl, wget or sh globally.

INVESTIGATION PIVOTS

Host
Role, exposure, firmware and sensor health.

Command
Full URL and syntax variants.

Parent
Cron daemon, init, profile or operator shell.

Filesystem
Crontab, unit, init and rc.local changes.

Network
Loader destination and encrypted TCP/443 C2.

Payload
Architecture, hash and execution lineage.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Bootstrap installer	An approved installer may pipe a remote script to a shell.	Require trusted URL, owner, package and change record.
Configuration management	Provisioning may use dual downloaders for portability.	Allowlist immutable script identity and managed parent.
Security lab replay	Analysts may reproduce the published command.	Require isolated asset and approved test window.

Baseline plan

- Search the full available DeviceProcessEvents retention.
- Inspect every matching raw event and process tree.
- Record owner, parent, URL and destination for legitimate matches.
- Do not add recurrence or rarity thresholds before measurement.

Validation procedure

- Confirm supported Linux onboarding and sensor health.
- Execute a safe controlled source-pattern command.
- Run the three split negative cases.
- Correlate the positive event with cron and network ground truth.

TEST CASES

CASE	EXPECTED RESULT
Run the complete pipeline against a controlled script	Expected match with complete fields.
Remove the curl fallback	No match.
Remove the pipe to /bin/sh	No match.
Remove output redirection	No match.

Hunting

Initial query

Candidate

Current status

Production

Measured precision

Optimized

Stable feedback

Success

The controlled full pattern matches with complete event identity, negative cases do not, and historical collisions are absent or explainable.

Rejection

Command-line capture is incomplete or approved software frequently reproduces the full pattern without a durable discriminator.

Failure

A controlled matching execution produces no usable DeviceProcessEvents record.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1059.004 · T1053.003	Unix Shell · Cron	Directly covers the shell pipeline installed by the source-observed cron persistence mechanism.

Detection coverage summary

Strong for the exact command pattern on onboarded Linux servers; weak for syntax changes, direct binary execution and unmanaged edge devices.

Residual risk

The main victim class may lack EDR. EvooolBot can change URLs, command syntax or persistence method; NDR, IPS, firmware inventory and incident response remain necessary.

ANALYST CONCLUSION

A selective hunt, not a substitute for edge-device coverage.

FortiGuard directly documents the persistence command and Microsoft documents the required process schema. This supports a Medium-confidence Hunting query. Coverage gaps, artifact mutability and lack of target-environment testing prevent Production classification.

REFERENCES

- FortiGuard Labs, “Multi-Functional Linux Botnet EvooolBot,” 13 August 2026; targeting observed since July 2026. <https://www.fortinet.com/blog/threat-research/multi-functional-linux-botnet-evoolbot>
- Microsoft Learn, “DeviceProcessEvents table,” updated 7 August 2026; accessed 16 August 2026. <https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table>
- Microsoft Learn, “Prerequisites for Microsoft Defender for Endpoint on Linux,” updated 12 June 2026; accessed 16 August 2026. <https://learn.microsoft.com/en-us/defender-endpoint/mde-linux-prerequisites>
- MITRE ATT&CK, T1059.004 and T1053.003. <https://attack.mitre.org/techniques/T1059/004/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	16 August 2026	Hunting	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.